

GeoNexus-RSD: Hierarchy- and Context-Aware Vision-Language Prompting for Oriented Remote Sensing Detection

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Abstract

Oriented object detection in remote sensing images must distinguish small, dense, and semantically similar targets under large scale variation and arbitrary rotation. We study GeoNexus-RSD, a detector-centered vision-language prompting framework that augments a strong oriented detector with taxonomy-aware class prompts, hierarchy regularization, scene/context adaptation, and conservative pseudo-label purification. The current route is DOTA2-first: DOTA2 is the primary benchmark, DIOR-R is the required cross-dataset validation, and FAIR1M is reserved for later fine-grained evidence. As of 2026-06-13, the completed GeoNexus DOTA2 S1 run reaches 0.6177 mAP and 0.6180 AP50 on `DOTA2_1024_500/ss_val`, improving over the matched RoI Transformer S0 baseline of 0.6088/0.6090. Seven DOTA2 S2 loss-0 runs show repeatable best-checkpoint signal with mean 0.620606, but final checkpoints average 0.616655 and remain unstable. On sanitized DIOR-R, RoI Transformer S0 reaches 0.6531/0.6530 and two S1 replicas reach 0.6751/0.675 and 0.6690/0.669, while S2 hierarchy replicas have clean startup but pending metrics. The draft therefore reports current evidence conservatively: measured DOTA2 and DIOR-R results are separated from active, planned, paused, and archive-only evidence.

1 Introduction

Remote sensing object detection supports transportation monitoring, maritime surveillance, urban analytics, disaster assessment, and defense-oriented interpretation. Compared with natural-image detection, aerial scenes contain very large images, dense object layouts, long-tail categories, strong scale variation, and objects with arbitrary orientation. DOTA made rotated object detection a standard benchmark for such imagery [1]. Strong oriented detectors such as RoI Transformer, Oriented R-CNN, ReDet, and Oriented RepPoints improve the geometric path through transformed RoIs, oriented proposals, rotation-equivariant features, or point-set representations [2, 3, 4, 5].

Most oriented detectors still treat categories as independent closed-set labels. This is limiting for remote sensing. Fine-grained targets share local appearance, class definitions vary across datasets, and the same visual crop may need surrounding context to disambiguate. A bright elongated region may be a small vehicle, a ship, a harbor structure, or part of an airport scene depending on nearby roads, water, or facilities. Language-aligned models provide a path for adding semantic priors without abandoning standard detector protocols.

GeoNexus-RSD keeps the detector backbone and oriented-box protocol central, then injects structured semantic evidence through prompts. The core modules are:

- a hierarchical prompt bank grounded in class names, aliases, parent categories, and confusing class pairs;
- a prompt-conditioned classification path using real RemoteCLIP text embeddings where available;
- a hierarchy regularizer that penalizes semantically inconsistent class scores;
- a scene/context adapter and pseudo-label purification stage, both kept paused until DOTA2 S1/S2 and DIOR-R become stable.

This draft is not a final submission. It is a stronger paper draft and evidence ledger for the DOTA2-first route. Its contribution is to define the method, record the present measured evidence, show publication-style figures and tables, and make unsupported items explicit.

2 Related Work

Oriented remote sensing detection. DOTA established a large-scale aerial object benchmark with oriented boxes and dense object layouts [1]. RoI Transformer learns rotated RoIs from horizontal proposals [2]. Oriented R-CNN simplifies the two-stage oriented detection pipeline with oriented proposals [3]. ReDet introduces rotation-equivariant feature extraction for aerial scenes [4]. Oriented RepPoints models oriented objects through adaptive point sets [5]. GeoNexus-RSD follows this detector family and asks whether structured language evidence can improve classification behavior while preserving the same oriented-box evaluation protocol.

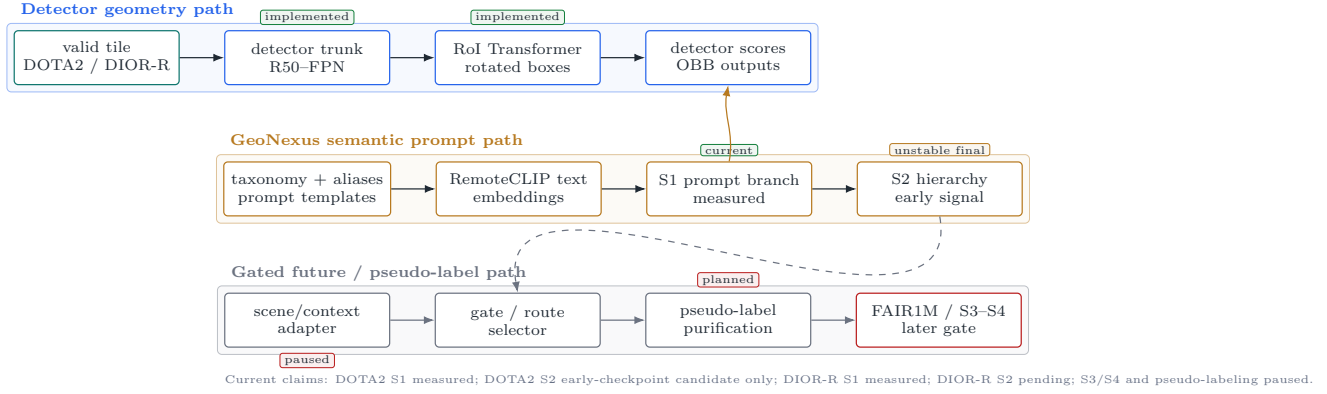


Figure 1: GeoNexus-RSD detector/prompt architecture. The detector geometry path and S1 prompt path are implemented paper-facing modules; S2 is an early-checkpoint candidate with unstable finals; context gating and pseudo-label purification remain paused or planned until the DOTA2 and DIOR-R gates are stable.

Vision-language detection and remote sensing prompts. CLIP-style pretraining aligns visual and language representations at scale [11]. GLIP, RegionCLIP, and Detic extend language supervision to detection [12, 13, 14]. Remote sensing transfer is difficult because aerial targets are small, rotated, and often context-dependent. RemoteCLIP, SkyScript, and GeoRSCLIP motivate remote-sensing-specific image-text alignment [7, 8, 9]. OpenRSD is the closest reference direction for open-prompt remote sensing detection [6]. GeoNexus-RSD uses this literature to motivate prompt structure but does not claim open-vocabulary detection unless a real open-vocabulary or vocabulary-robustness evaluation is run.

Pseudo-label denoising. Unbiased Teacher and Soft Teacher show that pseudo labels can improve detection when noise is controlled [15, 16]. Remote sensing pseudo labels are fragile because small objects, dense duplicates, and ambiguous categories create false positives and class swaps. SemiEarth motivates vision-language purification in remote sensing [10]. GeoNexus-RSD adapts this idea to oriented detection only as a planned stage until DOTA2 S1/S2 and DIOR-R are stable.

3 Method

Given an aerial tile I , an oriented detector predicts rotated boxes $b_i = (x_i, y_i, w_i, h_i, \theta_i)$ and class scores over vocabulary $\mathcal{C}$. GeoNexus-RSD keeps the detector-native geometric branch and adds a semantic branch based on a taxonomy $\mathcal{T}$ over $\mathcal{C}$. Figure 1 shows the detector-style architecture.

3.1 Hierarchical Prompt Bank

For each class $c \in \mathcal{C}$, the prompt bank stores class names, aliases, parent descriptors, scene descriptors, and contrastive phrases:

$$\mathcal{P}_c = \{p_{c,1}, \dots, p_{c,K}\}, \quad e_{c,k} = T(p_{c,k}). \quad (1)$$

The class embedding is a weighted prompt mixture:

$$\bar{e}_c = \sum_{k=1}^K \alpha_{c,k} e_{c,k}, \quad \sum_{k=1}^K \alpha_{c,k} = 1. \quad (2)$$

In the current DOTA2 and DIOR-R routes, RemoteCLIP ViT-B/32 provides real text embeddings. The DOTA2 S2 prompt artifact contains 18 finite, normalized class embeddings and an 18×18 relation matrix; the DIOR-R S2 artifact contains 20 finite, normalized class embeddings and a 20×20 relation matrix. This matters because the evidence should not be described as VLM-based if a run falls back to deterministic hash embeddings.

3.2 Hierarchy Regularization

The taxonomy supplies parent-child and related-category edges $\mathcal{E}_{\mathcal{T}}$. A simple margin regularizer penalizes inconsistent score orderings:

$$\mathcal{L}_{hier} = \sum_{(u,v) \in \mathcal{E}_{\mathcal{T}}} \max(0, s_v - s_u + m), \quad (3)$$

Table 1: DOTA2 evidence after the June 13 S2 stabilization analysis.

Run	Status	mAP	AP50	Use
DOTA2 S0 RoI Transformer	complete	0.6088	0.6090	matched valid-PNG detector baseline
DOTA2 S1 RemoteCLIP	complete	0.6177	0.6180	paper-facing positive DOTA2 result
DOTA2 S2 loss-0 best mean	candidate	0.620606	-	all-seven best-checkpoint mean; early only
DOTA2 S2 loss-0 final mean	unstable	0.616655	-	all-seven final mean; below S1

where s_u and s_v are prompt-conditioned scores and m is a margin. The term is designed for fine-grained aerial classes where errors often occur inside semantically related groups such as vehicle-like, ship/harbor, or airport/helipad categories. The DOTA2 S2 loss-0 stabilization runs show a best-checkpoint mean above S1 but final-checkpoint instability, so the present draft treats S2 as an early-checkpoint candidate rather than a final improvement. DIOR-R S2 hierarchy replicas have clean startup and pending metrics.

3.3 Scene Context Adapter

Let $F(I)$ be the detector feature map, $g = \text{GAP}(F(I))$ a tile descriptor, and r_i a region feature. The context-adapted class embedding is:

$$\tilde{e}_{c,i} = \bar{e}_c + A([g, r_i, \bar{e}_c]), \quad (4)$$

where $A(\cdot)$ is a lightweight adapter. The prompt-conditioned classification logit is:

$$z_{i,c} = \tau^{-1} \frac{\phi(r_i)^\top \psi(\tilde{e}_{c,i})}{\|\phi(r_i)\|_2 \|\psi(\tilde{e}_{c,i})\|_2}. \quad (5)$$

The box regression branch remains detector-native. This keeps localization comparable to standard oriented baselines. The context adapter is a core method component, but it remains paused as a paper-facing experiment until DOTA2 S2 and DIOR-R are stable.

3.4 Pseudo-Label Purification

For a candidate pseudo box $\hat{b}_j$ with class $\hat{c}_j$, the planned purification score is:

$$q_j = \sigma(\beta_1 d_j + \beta_2 h_j + \beta_3 v_j + \beta_4 a_j), \quad (6)$$

where d_j is detector confidence, h_j hierarchy consistency, v_j crop-text agreement, and a_j geometric plausibility. Accepted pseudo labels are weighted softly:

$$\mathcal{L} = \mathcal{L}_{det}^{sup} + \lambda_h \mathcal{L}_{hier} + \lambda_p \sum_j q_j \left(\mathcal{L}_{cls}^{pseudo}(j) + \lambda_b \mathcal{L}_{box}^{pseudo}(j) \right). \quad (7)$$

This stage is part of the intended GeoNexus-RSD design but is not yet completed evidence.

4 Experimental Route and Evidence

4.1 DOTA2-First Route

DOTA2 is the primary benchmark. The current matched S0 detector is RoI Transformer on DOTA2_1024_500/ss_val after valid-PNG recovery. The paper-facing positive result is one completed GeoNexus S1 run initialized with RemoteCLIP prompts. Figure 2 separates measured DOTA2 and DIOR-R evidence from unstable, pending, paused, and archive-only evidence.

The headline DOTA2 interpretation is narrow. GeoNexus S1 GPU-1 improves over the matched RoI Transformer S0 baseline by 0.0089 mAP and 0.0090 AP50. The S2 loss-0 stabilization set is stronger when selecting best checkpoints, with all-seven best mean 0.620606, but final checkpoints average 0.616655 and fall below S1. Therefore S2 supports an early-checkpoint candidate, not a completed final-epoch stability claim.

Experiment route status on 2026-06-13



Figure 2: Experiment route/status figure. DOTA2 and DIOR-R measured results are separated from unstable S2 finals, pending DIOR-R S2 replicas, archive-only DOTA v1.5 diagnostics, and paused S3/S4, pseudo-label, routing, and FAIR1M work.

Table 2: Route status used to prevent overclaiming active or paused modules.

Track	Role	Status
DOTA2	primary benchmark	S0/S1 measured; S2 best-checkpoint candidate
DIOR-R	required validation	S0/S1 measured; S2 metrics pending
DOTA v1.5	archive only	diagnostic history, not headline evidence
S3/S4	paused	resume only after S2 and DIOR-R gates are stable
Pseudo-label / FAIR1M	planned	no paper-facing metric claim yet

4.2 Class-Level and Visual Evidence

The current draft includes class-level S0 evidence as appendix-style diagnostic context. Real qualitative detections and accepted/rejected pseudo-label examples must be added only after valid inference artifacts are archived.

4.3 GeoNexus Stage Ablation Status

Table 2 intentionally mixes measured route points with pending/paused rows, but each row is marked by role and status. DOTA v1.5 rows are archive diagnostics and must not be used as headline benchmark evidence after the 2026-06-06 pivot.

4.4 DIOR-R Diagnostic Gate

DIOR-R is required for cross-dataset validation. The earlier NaN/Inf detector path was addressed through sanitized-label scans and bounded train-step diagnostics. The current sanitized RoI Transformer S0 reaches 0.6531/0.6530, and RemoteCLIP S1 replicas reach 0.6751/0.675 and 0.6690/0.669. DIOR-R S2 hierarchy replicas have clean startup, but their metrics are pending and must not be reported as gains.

4.5 Prompt and Reporting Tables

The final paper must still add measured parameters, memory, latency, and FLOPs, or omit efficiency claims entirely.

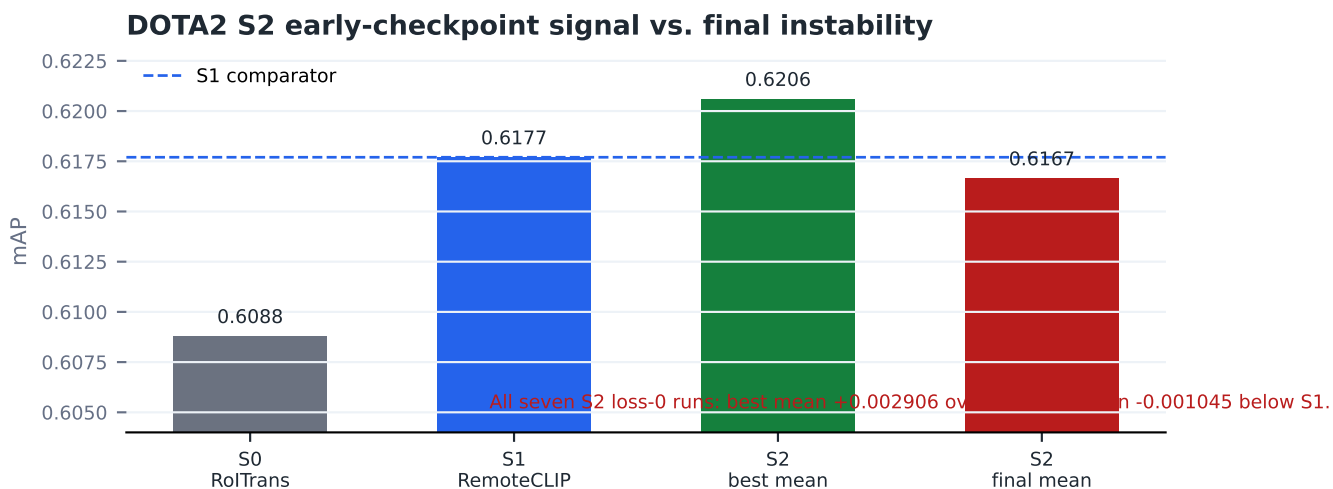


Figure 3: DOTA2 S2 stability chart. S2 best checkpoints are above S1 on average, but final checkpoints remain unstable and should not be cited as final S2 evidence.

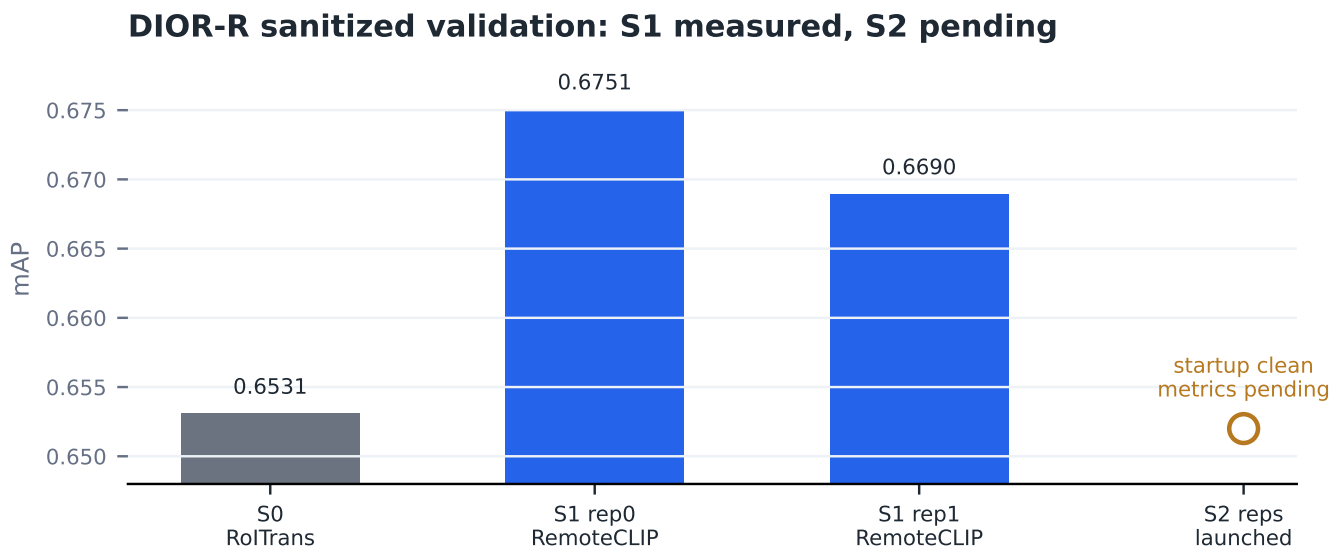


Figure 4: DIOR-R sanitized transfer evidence. RoI Transformer S0 is the current DIOR-R detector baseline, two RemoteCLIP S1 replicas are measured, and S2 hierarchy replicas are launched with clean startup but pending metrics.

4.6 Archive-Only DOTA v1.5 Diagnostics

DOTA v1.5 GeoNexus experiments exercised the hierarchy and context code paths, but they are no longer the formal benchmark route. They should remain in experiment records and may be used for debugging narratives, not as headline table evidence. Any final paper table must keep DOTA v1.5 separate from DOTA2 and DIOR-R.

5 Required Final Analyses

The final submission should contain only measured rows. The current draft records the required analysis set so that future work can close the evidence gaps without shifting the story.

All tables must record dataset version, split, config, checkpoint, metric source, and whether embeddings are real VLM embeddings or fallback embeddings. DOTA2 remains the primary benchmark. DIOR-R is required for cross-dataset validation. FAIR1M is a stretch dataset for fine-grained hierarchy analysis only after DOTA2 and DIOR-R are stable.

DOTA2 S0 class-level diagnostic evidence

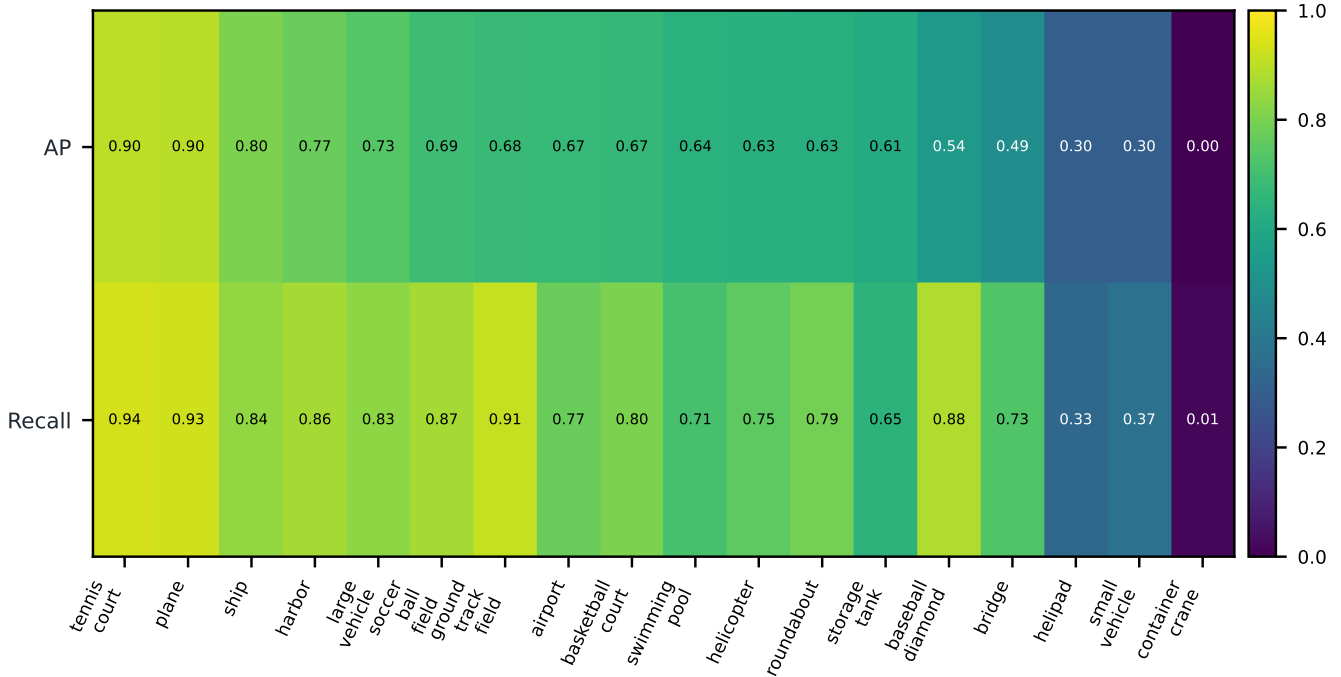


Figure 5: DOTA2 S0 class evidence groups generated from the RoI Transformer valid-PNG metric record. This is diagnostic context for prompt design and confusing-class analysis.

Table 3: DIOR-R sanitized route evidence and pending S2 status.

Run	Status	mAP	AP50	Use
DIOR-R S0 RoI Transformer	sanitized S0	0.6531	0.6530	current S0 leader
DIOR-R S1 rep0	complete	0.6751	0.675	stronger S1 replica; source for S2
DIOR-R S1 rep1	complete	0.6690	0.669	stability replica
DIOR-R S2 hierarchy reps	launched	pending	pending	clean startup; no metric claim

6 Limitations and Next Steps

The current evidence has four limitations. First, the positive DOTA2 S1 result is one completed main run, and S2 final checkpoints remain unstable despite repeatable best-checkpoint signal. Second, DIOR-R S1 is now measured on sanitized labels, but DIOR-R S2 metrics are pending. Third, the context adapter, pseudo-label purification, and FAIR1M evidence remain paused. Fourth, no measured pseudo-label quality or efficiency table is available.

The immediate next steps are:

1. decide a defensible DOTA2 S2 checkpoint-selection protocol or archive S2 as unstable;
2. finish and archive DIOR-R S2 hierarchy replicas;
3. compare DIOR-R S0, S1, and S2 best/final checkpoints separately;
4. keep DOTA v1.5 diagnostics out of headline benchmark tables;
5. measure efficiency and add valid qualitative detections only from archived inference artifacts.

Only after these gates should the work advance to scene adaptation, pseudo-label purification, routing, or FAIR1M.

7 Conclusion

GeoNexus-RSD studies hierarchy- and context-aware vision-language prompting for oriented remote sensing detection. The current DOTA2-first evidence shows one completed GeoNexus S1 run above the matched RoI Transformer baseline, DOTA2 S2 best-checkpoint signal with unstable finals, and measured DIOR-R S1 replicas above the sanitized RoI Transformer S0 baseline. This draft therefore frames the method and evidence conservatively: DOTA2 is the primary benchmark, DIOR-R is the required

Table 4: Source map for the June 13 generated paper assets.

Claim	Value	Local source
DOTA2 S0	0.6088 / 0.6090	docs/experiments/20260603_s0_dota2_roi_trans_validpng_metrics.json
DOTA2 S1	0.6177 / 0.6180	docs/experiments/20260608_dota2_s1_complete_and_s2_launch.md
DOTA2 S2 aggregate	best 0.620606; final 0.616655	docs/experiments/20260611_dota2_s2_loss0_replicates_analysis.md
DIOR-R S0	0.6531 / 0.6530	docs/experiments/20260613_dior_r_s0_sanitized_long_interim.md
DIOR-R S1 replicas	0.6751 / 0.675; 0.6690 / 0.669	docs/experiments/20260613_dior_r_geonexus_s1_s0e52_replicas_metrics.json
DIOR-R S2	startup clean; metrics pending	docs/experiments/20260613_dior_r_geonexus_s2_hierarchy_replicas_launch.md

Table 5: Archive-only DOTA v1.5 reduced tiled diagnostics. These values are not headline evidence.

Stage	Setting	Best	Final
S0	RoI Transformer 3x baseline	0.2644 / 0.2640	0.2612 / 0.2610
S1	RemoteCLIP prompt head	0.2666 / 0.2670	0.2506 / 0.2510
S2	Hierarchy regularizer 72e	0.3757 / 0.3760	0.3738 / 0.3740
S3	Scene adapter 72e	0.3800 / 0.3800	0.3759 / 0.3760

validation gate, DOTA v1.5 is archive-only, and S2, pseudo-label, S3/S4, and FAIR1M outcomes are not treated as completed final evidence.

Table 6: Required final analyses. Planned rows must be replaced by measured values before submission.

Analysis	Required datasets	Purpose
Main comparison	DOTA2, DIOR-R	Compare S0 detector and GeoNexus variants
Core ablation	DOTA2, DIOR-R	Isolate prompt bank, hierarchy, and context modules
Prompt robustness	DOTA2	Compare labels, aliases, parents, and mixed prompts
Pseudo-label quality	DOTA2, DIOR-R when stable	Measure accepted/rejected pseudo-label precision
Efficiency	DOTA2	Report parameters, memory, latency, and FLOPs per tile
Qualitative evidence	DOTA2, DIOR-R	Show valid detections and failure cases from archived inference artifacts

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